



Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Swetha Nagasri
Student Name	Myadagoni Ramya Sree
Roll Number	2520030094
Branch	CSE
Course Outcome	CO-5

Case Study Topic : Snapdeal Product Catalog Sorting using TimSort

Work Done Summary :

In this case study, TimSort Algorithm was implemented to sort product ratings in Snapdeal's catalog management system. TimSort is a hybrid sorting algorithm derived from Merge Sort and Insertion Sort and is used internally by Java's Arrays.sort() for object sorting.

The product ratings were processed as partially sorted data. TimSort first identified naturally occurring sorted runs and then used insertion sort to extend short runs. Finally, merge operations combined the sorted runs into a completely ordered list.

This approach provides excellent performance on real-world datasets that already contain partially sorted sequences.

Implementation Details:

1. Represented product ratings using an integer array.
2. Identified natural ascending and descending runs.
3. Reversed descending runs to maintain ascending order.
4. Extended short runs using insertion sort.
5. Stored runs on a stack structure.
6. Applied TimSort merge rules to maintain balance.
7. Merged adjacent sorted runs efficiently.
8. Generated the final sorted catalog list.
9. Measured sorting performance on partially sorted data.

Result: Output Screenshots:

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SNAPDEAL PRODUCT CATALOG SORTING
Using TimSort Algorithm
=====

Input Array:

3 7 11 5 4 6 9 12 1 2 8 10 13 14

Detecting Natural Runs...

Run 1 Found:
3 7 11

Run 2 Found:
5 4

Descending Run Detected
Reversing Run...

Updated Run:
4 5

Run 3 Found:
6 9 12

Run 4 Found:
1 2

Run 5 Found:
8 10 13 14
```

```
=====
RUN STACK CREATED
=====

Run 1 : [3, 7, 11]
Run 2 : [4, 5]
Run 3 : [6, 9, 12]
Run 4 : [1, 2]
Run 5 : [8, 10, 13, 14]

=====
MERGING RUNS
=====

Merge Run1 and Run2

Result:
3 4 5 7 11

Merge Run3

Result:
3 4 5 6 7 9 11 12

Merge Run4

Result:
1 2 3 4 5 6 7 9 11 12

Merge Run5

Result:
1 2 3 4 5 6 7 8 9 10 11 12 13 14

=====
FINAL SORTED ARRAY
=====

1 2 3 4 5 6 7 8 9 10 11 12 13 14

=====
SORTING COMPLETED SUCCESSFULLY
=====
```

Conclusion :

The TimSort algorithm successfully sorted the Snapdeal product rating catalog. By exploiting existing ordered sequences in the data, TimSort reduced unnecessary comparisons and improved performance. The algorithm efficiently combined insertion sort and merge sort techniques, making it highly suitable for real-world applications involving partially sorted datasets such as e-commerce product catalogs.

GitHub Link:

Student Signature	Faculty Signature

